PADHAKU
Contemporary India · Class 10 Social Science



CHAPTER 5

Minerals and Energy Resources

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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🤔 Let's think · Metals in daily life

? **Can you list a number of items used in your house made of metals? Where do these metals come from?**

💡 SAMPLE ANSWER

Items made of metals:

- Kitchen: steel utensils, pressure cooker, aluminium pans, spoons, gas stove
- House: iron almirah, door locks, hinges, nails, taps, window grills
- Electrical: copper wires, fans, switches, bulbs
- Others: bicycle, scissors, keys, coins

Where they come from: metals are extracted from **minerals** found in the rocks of the earth's crust. The rock containing the mineral (ore) is mined and the metal is taken out after proper refinement, e.g. iron from hematite/magnetite, aluminium from bauxite, copper from copper ore.

? **Toothpaste: abrasive minerals (silica, limestone, aluminium oxide, phosphates), fluorite, rutile/ilmenite/anatase and mica are used. Find out where these minerals are found.**

💡 SAMPLE ANSWER

- **Limestone:** sedimentary rocks; found in many states, e.g. Rajasthan, Madhya Pradesh, Andhra Pradesh, Chhattisgarh
- **Aluminium oxide:** from bauxite: Odisha (Panchpatmali, Koraput), Amarkantak plateau, Maikal hills, Bilaspur-Katni
- **Mica:** Koderma-Gaya-Hazaribagh belt (Jharkhand), around Ajmer (Rajasthan), Nellore (Andhra Pradesh)
- **Silica:** quartz sand, found widely in river and sea sands
- **Rutile and ilmenite:** beach sands of Kerala, Tamil Nadu and Odisha
- **Fluorite:** mainly Gujarat and Rajasthan
- **Phosphate minerals:** rock phosphate of Rajasthan (Udaipur) and Madhya Pradesh

? **Dig a little deeper and find out how many minerals are used to make a light bulb?**

💡 **SAMPLE ANSWER**

Many minerals are used in one bulb, for example:

- **Glass bulb:** silica sand, soda ash and limestone
- **Filament:** tungsten (old bulbs); LED bulbs use gallium and other elements in the chip
- **Base and wires:** aluminium or brass (copper + zinc), copper, nickel, lead (solder)
- **Inside gas:** argon (from air)

So about **8–10 minerals** go into making a simple light bulb.

? Dig a little deeper and collect "Nutritional Facts" printed on food labels.

💡 SAMPLE ANSWER

Sample (from a biscuit / milk packet label):

Nutrient	Per 100 g
Energy	450 kcal
Protein	7 g
Carbohydrate	70 g
Fat	15 g
Calcium	120 mg
Iron	4 mg
Sodium	300 mg

Minerals on labels are usually **calcium, iron, sodium, potassium, zinc and magnesium**. Though needed in small amounts, they are essential for life processes.

? Minerals are found in varied forms, from the hardest diamond to the softest talc. Why are they so varied?

✔ ANSWER

- A mineral forms from a certain combination of elements.

- Which mineral forms depends on the **physical and chemical conditions** (heat, pressure, etc.) under which the material forms.
- This gives minerals a wide range of **colours, hardness, crystal forms, lustre and density**.

⭐ Dig a little deeper · Rat-Hole Mining

? What is the difference between an open pit mine, a quarry and an underground mine with shafts?

💡 SAMPLE ANSWER

- **Open pit mine:** a large, deep pit dug from the surface to take out minerals lying near the surface, e.g. coal, iron ore, bauxite.
- **Quarry:** a shallow open-surface pit from which building stones, limestone, marble, sand or gravel are cut and taken out.
- **Underground (shaft) mine:** used when the mineral lies deep; vertical shafts and tunnels are dug to reach it, e.g. deep coal and gold mines. It is costlier and riskier.

⭐ Dig a little deeper · Ferrous Minerals

? Superimpose the maps showing distribution of iron ore, manganese, coal and iron and

steel industry. Do you see any correlation? Why?

✔ ANSWER

Yes, there is a close correlation.

- Iron ore, manganese and coal are all found together in the **Chota Nagpur plateau** region (Jharkhand, Odisha, Chhattisgarh, West Bengal).
- Most iron and steel plants (Jamshedpur, Bokaro, Durgapur, Rourkela, Bhilai) are in this same region.

Why: these are the main raw materials of steel. They are bulky and heavy, so plants are set up near them to **cut transport costs**.

⭐ Dig a little deeper · Bauxite

? **Locate the mines of Bauxite on the physical map of India.**

💡 SAMPLE ANSWER

Mark these on the map:

- **Panchpatmali** deposits, Koraput district (Odisha): most important
- **Amarkantak plateau** (Madhya Pradesh–Chhattisgarh border)
- **Maikal hills** (Madhya Pradesh–Chhattisgarh)
- **Bilaspur-Katni** plateau region

(Use the map 'India: Distribution of Iron Ore, Manganese, Bauxite and Mica' in your book.)

⭐ Dig a little deeper · Rock Minerals

? Study the maps to explain why Chota Nagpur is a storehouse of minerals.

✔ ANSWER

- The Chota Nagpur plateau is made of very old **peninsular (igneous and metamorphic) rocks**, which hold most metallic minerals, coal and mica.
- **Iron ore:** Singhbhum (Gua, Noamundi) and nearby Odisha belt.
- **Coal:** Gondwana coal of the Damodar valley: Jharia, Raniganj, Bokaro.
- **Mica:** Koderma-Gaya-Hazaribagh belt.
- **Copper:** Singhbhum; also manganese and bauxite nearby.

So many minerals are found here together; that is why it is called the storehouse of minerals.

⭐ Hazards of Mining

? Have you ever wondered about the efforts the miners make in making life comfortable for you? What are the impacts of mining on the health of the miners and the environment?

✔ ANSWER

On miners' health:

- Dust and noxious fumes cause **pulmonary (lung) diseases**.
- Constant threat of **collapsing mine roofs, inundation (flooding) and fires** in coal mines.

On the environment:

- Water sources in the region get **contaminated**.
- Dumping of waste and slurry **degrades land and soil**.
- Stream and river pollution increases; dust pollutes the air.

Stricter safety rules and environmental laws are needed so that mining does not become a "killer industry".

⭐ Dig a little deeper · Conservation of Minerals

? Make a list of items where substitutes are being used instead of minerals. Where are these substitutes obtained from?

💡 SAMPLE ANSWER

Item	Mineral earlier	Substitute	Source of substitute
Buckets, pipes, chairs	Iron, steel	Plastic	Petroleum
Telephone cables	Copper	Optical fibre	Glass (silica)
Car body parts	Steel	Fibreglass, plastics	Glass, petroleum
Water tanks	Iron, cement	PVC / plastic	Petroleum
Packaging	Tin, aluminium	Paper, cardboard	Wood (forests)
Utensils	Metals	Bamboo, clay, plant-fibre ware	Plants, soil

🧪 Activity · Natural Gas

? Collect information about cross country natural gas pipelines laid by GAIL (India) under "One Nation One Grid".

💡 SAMPLE ANSWER

- **Hazira-Vijaipur-Jagdishpur (HVJ):** the first 1,700 km cross-country pipeline; linked Mumbai High and Bassein gas fields with fertiliser, power and industrial complexes of western and northern India.
- **Dahej-Vijaipur** and **Vijaipur-Dadri** pipelines extend this network.
- **Jagdishpur-Haldia-Bokaro-Dhamra (Pradhan Mantri Urja Ganga):** takes gas to eastern India.
- **Kochi-Koottanad-Bengaluru-Mangaluru:** serves Kerala and Karnataka.
- **Barauni-Guwahati:** links the North-Eastern states to the grid.

Overall the network has grown from 1,700 km to about **18,500 km** and is expected to reach **34,000 km** as a national Gas Grid.

⭐ Petroleum

? From the map locate the 3 major off shore fields of western India.

💡 SAMPLE ANSWER

- **Mumbai High** (in the Arabian Sea, off Mumbai)
- **Bassein** (Bassien), north of Mumbai High
- **Aliabet**, in the Gulf of Khambhat (Gujarat)

(Check the map 'India: Distribution of Coal, Oil and Natural Gas' in your book.)

⭐ Electricity

? Name some river valley projects and write the names of the dams built on these rivers.

✔ ANSWER

River valley project	River	Dam
Bhakra Nangal	Satluj	Bhakra and Nangal dams
Damodar Valley Corporation	Damodar	Tilaiya, Konar, Maithon, Panchet
Hirakud	Mahanadi	Hirakud dam
Nagarjuna Sagar	Krishna	Nagarjuna Sagar dam
Sardar Sarovar	Narmada	Sardar Sarovar dam
Tehri	Bhagirathi	Tehri dam
Rihand	Rihand (Son)	Govind Ballabh Pant Sagar
Kopili Hydel Project	Kopili	Khandong and Umrangso dams (Assam)

Activity **Collect information about thermal/hydel power plants located in your state. Show them on the map of India.**

💡 SAMPLE ANSWER

Sample (Delhi / Uttar Pradesh):

- **Thermal:** Badarpur (Delhi), Singrauli, Rihand, Panki, Harduaganj, Parichha (Uttar Pradesh)

- **Hydel:** Rihand hydel project (Uttar Pradesh), Tehri (Uttarakhand)

Mark thermal plants with a dot and hydel plants with a different sign, as in the book's map. Write about your own state in the same way.

⭐ Non-Conventional Sources of Energy

? Locate the 6 nuclear power stations and find out the state in which they are located.

✔ ANSWER

Nuclear power station	State
Narora	Uttar Pradesh
Rawat Bhata	Rajasthan
Kakrapara	Gujarat
Tarapur	Maharashtra
Kaiga	Karnataka
Kalpakkam	Tamil Nadu

Activity **Collect information about newly established solar power plants in India.**

💡 SAMPLE ANSWER

- **Bhadla Solar Park**, Jodhpur (Rajasthan): one of the largest solar parks in the world
- **Pavagada Solar Park**, Tumakuru (Karnataka)
- **Kurnool Ultra Mega Solar Park** (Andhra Pradesh)

- **Rewa Ultra Mega Solar Project** (Madhya Pradesh): supplies power to Delhi Metro
- **Kamuthi Solar Power Project** (Tamil Nadu)

These reduce the use of coal and help supply clean electricity.

📝 Exercises

1 (i) **Which one of the following minerals is formed by decomposition of rocks, leaving a residual mass of weathered material? (a) coal (b) bauxite (c) gold (d) zinc**

✔ ANSWER

(b) bauxite

It forms when surface rocks decompose and soluble parts are removed, leaving a residual mass.

1 (ii) **Koderma, in Jharkhand is the leading producer of which one of the following minerals? (a) bauxite (b) mica (c) iron ore (d) copper**

✔ ANSWER

(b) mica

The Koderma-Gaya-Hazaribagh belt of Jharkhand is the leading mica producer.

1 (iii) **Minerals are deposited and accumulated in the stratas of which of the following rocks? (a) sedimentary rocks (b) metamorphic rocks (c) igneous rocks (d) none of the above**

✔ ANSWER

(a) sedimentary rocks

In sedimentary rocks minerals occur in beds or layers (horizontal strata).

1 (iv) **Which one of the following minerals is contained in the Monazite sand? (a) oil (b) uranium (c) thorium (d) coal**

✔ ANSWER

(c) thorium

The Monazite sands of Kerala are rich in thorium.

2 (i) (a) **Distinguish between ferrous and non-ferrous minerals.**

✔ ANSWER

Ferrous minerals	Non-ferrous minerals
Contain iron	Do not contain iron
E.g. iron ore, manganese, nickel, cobalt	E.g. copper, bauxite, lead, zinc, gold

About three-fourths of the value of metallic mineral production; base of metallurgical industries	India's reserves and production are not very satisfactory

2 (i) (b) **Distinguish between conventional and non-conventional sources of energy.**

✔ ANSWER

Conventional sources	Non-conventional sources
In use for a long time	Developed recently
Mostly non-renewable and polluting (coal, petroleum)	Renewable and cleaner
E.g. firewood, cattle dung cake, coal, petroleum, natural gas, electricity	E.g. solar, wind, tidal, geothermal, biogas, atomic energy

2 (ii) **What is a mineral?**

✔ ANSWER

- Geologists define a mineral as a **"homogenous, naturally occurring substance with a definable internal structure"**.
- Minerals are found in varied forms, from the hardest diamond to the softest talc.
- Rocks are combinations of minerals.

2 (iii) **How are minerals formed in igneous and metamorphic rocks?**

✔ ANSWER

- Minerals occur in the **cracks, crevices, faults or joints** of these rocks.
- Minerals in liquid/molten and gaseous forms are forced upward through cavities towards the earth's surface, where they **cool and solidify**.
- Smaller occurrences are called **veins**, larger ones **lodes**, e.g. tin, copper, zinc and lead.

2 (iv) **Why do we need to conserve mineral resources?**

✔ ANSWER

- Minerals take **millions of years** to form; replenishment is infinitely slow compared to present consumption.
- So they are **finite and non-renewable**; workable deposits are only one per cent of the earth's crust.
- Continued extraction **raises costs** as ores come from greater depths and are of lower quality.

3 (i) **Describe the distribution of coal in India.**

✔ ANSWER

Coal is the most abundantly available fossil fuel in India and meets a large part of its energy needs. In India coal occurs in rock series of two main geological ages:

1. **Gondwana coal:** a little over 200 million years old; most of India's coal, including **metallurgical coal**.

2. Major Gondwana coal is in the **Damodar valley** (West Bengal–Jharkhand); **Jharia, Raniganj and Bokaro** are important coalfields.

3. The **Godavari, Mahanadi, Son and Wardha valleys** also contain coal deposits.

4. **Tertiary coal:** only about 55 million years old; found in the north-eastern states of **Meghalaya, Assam, Arunachal Pradesh and Nagaland**.

5. **Lignite** (low grade brown coal): main reserves at **Neyveli** in Tamil Nadu, used for generating electricity.

6. Coal is bulky and loses weight on use, so heavy industries and thermal power stations are located on or near the coalfields.

Thus most of India's coal is found in the peninsular plateau, especially the Damodar valley.

3 (ii) **Why do you think that solar energy has a bright future in India?**

✔ ANSWER

Solar energy has a bright future in India because:

1. India is a **tropical country** with an abundance of sunlight for most of the year, so it has enormous possibilities of tapping solar energy.

2. **Photovoltaic technology** converts sunlight directly into electricity.

3. It is a **renewable**, inexhaustible and clean source; it does not cause pollution like fossil fuels.

4. Rising prices and possible shortages of oil and gas make energy supply uncertain; solar energy reduces this **dependence on fossil fuels**.

5. It is fast becoming popular in **rural and remote areas**, where grid supply is poor.

6. Big solar power plants are being set up; they reduce rural households' dependence on **firewood and dung cakes**, which helps conserve the environment and keeps dung available as manure for agriculture.

Hence solar energy can meet India's growing energy needs in a sustainable way.

🧪 Activity · Crossword

Across **Fill the name of the correct mineral in the crossword (Across).**

✔ ANSWER

1. A ferrous mineral (9): **MANGANESE**

2. Raw material for cement industry (9): **LIMESTONE**

3. Finest iron ore with magnetic properties (9): **MAGNETITE**

4. Highest quality hard coal (10): **ANTHRACITE**

5. Aluminium is obtained from this ore (7): **BAUXITE**

6. Khetri mines are famous for this mineral (6): **COPPER**

7. Formed due to evaporation (6): **GYPSUM**

Down **Fill the name of the correct mineral in the crossword (Down).**

✔ ANSWER

1. Found in placer deposit (4): **GOLD**

2. Iron ore mined in Bailadila (8): **HEMATITE**

3. Indispensable for electrical industry (4): **MICA**

4. Geological age of coal found in north east India (8): **TERTIARY**

5. Formed in veins and lodes (3): **TIN**

🎉 **Chapter 5 done! Revise the answers once more tomorrow.**